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United States Patent	12415083
Kind Code	B2
Date of Patent	September 16, 2025
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Implantable arrays for providing tumor treating fields

Abstract

An apparatus for generating an electric field has a plurality of elongate elements, the plurality of elongate elements including at least a first elongate element and a second elongate element. Each elongate element of the plurality of elongate elements has a proximal end, an opposed distal end and at least one stimulation zone. The plurality of elongate elements are coupled together at their respective proximal ends. The plurality of elongate elements are selectively moveable about and between a retracted position and a deployed position. In the retracted position, the respective distal ends of the first and second elongate elements are spaced by a first distance, and in the deployed position, the respective distal ends of the first and second elongate elements are spaced by a second distance that is greater than the first distance.

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Appl. No.:	18/500598
Filed:	November 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240082593 A1	Mar. 14, 2024

Related U.S. Application Data

continuation parent-doc US 17336919 20210602 US 11839771 child-doc US 18500598
us-provisional-application US 63033511 20200602
us-provisional-application US 63033518 20200602

Publication Classification

Int. Cl.: A61N1/05 (20060101); A61N1/04 (20060101); A61N1/06 (20060101); A61N1/40 (20060101)

U.S. Cl.:

CPC A61N1/40 (20130101); A61N1/05 (20130101);

Field of Classification Search

CPC: A61N (1/40); A61N (1/05); A61N (1/0529); A61N (1/06); A61B (2018/00285); A61B (2018/147); A61B (5/24); A61B (5/293); A61B (5/6868); A61B (5/291); A61B (5/6882); A61B (5/0031); A61B (2018/00363); A61B (2018/00577); A61B (2562/125); A61B (18/1482); A61B (18/16); A61B (2090/376); A61B (2090/3966); A61B (2090/3983); A61B (2562/046); A61B (2562/164); A61B (2562/166); A61B (2562/222); A61B (5/263); A61B (5/271); A61B (5/283); A61B (5/686)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/336,919, filed Jun. 2, 2021, now U.S. Pat. No. 11,839,771 B2, which claims priority to and the benefit of the filing date of U.S. Provisional Patent Application No. 63/033,511, filed Jun. 2, 2020, and U.S. Provisional Patent Application No. 63/033,518, filed Jun. 2, 2020, each of which is incorporated herein by reference in its entirety.

FIELD

(1) This application relates generally to apparatuses and methods for providing tumor treating fields and, in particular, for apparatuses and methods for implanting electrodes within a patient for providing tumor treating fields.

BACKGROUND

(2) Tumor Treating Fields, or TTFields, are low intensity (e.g., 1-3 V/cm) alternating electrical fields within the intermediate frequency range (100-300 kHz). This non-invasive treatment targets solid tumors and is described in U.S. Pat. No. 7,565,205, which is incorporated herein by reference in its entirety. TTFields disrupt cell division through physical interactions with key molecules during mitosis. TTFields therapy is an approved mono-treatment for recurrent glioblastoma, and an approved combination therapy with chemotherapy for newly diagnosed patients. Conventionally, these electrical fields are induced non-invasively by transducer arrays (i.e., arrays of electrodes) placed directly on the patient's scalp. TTFields also appear to be beneficial for treating tumors in other parts of the body.

SUMMARY

(3) Described herein, in various aspects, is an apparatus for generating an electric field. The apparatus can have plurality of elongate elements. The plurality of elongate elements can have at least a first elongate element and a second elongate element. Each elongate element of the plurality of elongate elements can have a proximal end, an opposed distal end, and at least one stimulation zone. The plurality of elongate elements can be coupled together at their respective proximal ends. The plurality of elongate elements can be selectively moveable about and between a retracted position and a deployed position. In the retracted position, the respective distal ends of the first and second elongate elements can be spaced by a first distance. In the deployed position, the respective distal ends of the first and second elongate elements can be spaced by a second distance that is greater than the first distance.

(4) In other aspects, described herein is an apparatus having a bladder and a plurality of stimulation zones coupled to the bladder. The plurality of stimulation zones can include a first stimulation zone and a second stimulation zone. The plurality of stimulation zones can be configured to produce an electric field. Inflation of the bladder can cause the plurality of stimulation zones to deploy from a retracted configuration to a deployed configuration. When the plurality of stimulation zones are in the retracted configuration, the first and second stimulation zones are spaced by a first distance. When the plurality of stimulation zones are in the deployed configuration, the first and second stimulation zones are spaced by a second distance that is greater than the first distance.

(5) In other aspects, described herein is an apparatus having an inflatable body and a plurality of stimulation zones coupled to the inflatable body. The plurality of stimulation zones can be configured to produce an electric field. Inflation of the body can cause elongation of the body along a longitudinal axis of the body.

(6) In other aspects, described herein is an apparatus having a first body, a plurality of stimulation zones disposed along the first body, a second body, and an actuator. The first body can be telescopically coupled to the second body so that the first body is slidable with respect to the second body along a longitudinal axis of the apparatus. The plurality of stimulation zones can be configured to produce an electric field. The actuator can be configured to slide the first body with respect to the second body.

(7) In other aspects, described herein is a system having a guide and an electric field-generating

assembly. The guide can include an inflatable body having a longitudinal axis. The electric field-generating assembly can be configured for receipt within the inflatable body. The electric field-generating assembly can include a cable having a length and a plurality of stimulation zones disposed along the length of the cable. The plurality of stimulation zones are configured to produce an electric field.

(8) Methods of using the disclosed apparatuses and systems are also described.

(9) Additional advantages of the invention will be set forth in part in the description that follows, and in part will be obvious from the description, or may be learned by practice of the invention. The advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features of the preferred embodiments of the invention will become more apparent in the detailed description in which reference is made to the appended drawings wherein:

(2) FIG. 1A is a schematic side view of an implantable tumor treating field apparatus as disclosed herein, wherein the implantable tumor treating field apparatus comprises elongate elements that are in a deployed configuration. FIG. 1B is a schematic side view of the implantable tumor treating field apparatus as in FIG. 1A, with the elongate elements in a retracted configuration.

(3) FIG. 2A is a schematic side view of an implantable tumor treating field apparatus as disclosed herein, wherein the implantable tumor treating field apparatus comprises elongate elements that are in a deployed configuration. FIG. 2B is a cross section of the implantable tumor treating field apparatus as in FIG. 2A, taken at line 2B-2B. FIG. 2C is a schematic side view of the implantable tumor treating field apparatus as in FIG. 2A, with the elongate elements in a retracted configuration.

(4) FIG. 3 is a schematic side view of an implantable tumor treating field apparatus as disclosed herein, wherein the implantable tumor treating field apparatus comprises elongate elements that are in a deployed configuration.

(5) FIG. 4A is a schematic side view of an inflatable implantable tumor treating field apparatus as disclosed herein, wherein the apparatus is in the deflated configuration. FIG. 4B is a schematic side view of the implantable tumor treating field apparatus as in FIG. 4A, wherein the apparatus is in the inflated configuration.

(6) FIG. 5A is a schematic side view of an inflatable implantable tumor treating field apparatus as disclosed herein, wherein the apparatus is in the deflated configuration. FIG. 5B is a schematic side view of the implantable tumor treating field apparatus as in FIG. 5A, wherein the apparatus is in the inflated configuration.

(7) FIG. 6A is a schematic side view of an inflatable implantable tumor treating field apparatus as disclosed herein, wherein the apparatus is in the deflated configuration. FIG. 6B is a schematic side view of the implantable tumor treating field apparatus as in FIG. 6A, wherein the apparatus is in the inflated configuration.

(8) FIG. 7 is a schematic side view of an inflatable implantable tumor treating field apparatus as disclosed herein, wherein the apparatus comprises a plurality of inflatable portions.

(9) FIG. 8A is a schematic cross sectional view of an implantable telescoping treating field apparatus as disclosed herein, wherein the apparatus is in a compact configuration. FIG. 8B is a schematic cross sectional view of the implantable telescoping treating field apparatus as in FIG. 8A wherein the apparatus is in an extended configuration.

(10) FIG. 9 is a schematic of an inflatable implantable tumor treating field apparatus as disclosed herein, wherein the apparatus comprises a plurality of independently inflatable bodies for controlling its direction of elongation.

(11) FIG. 10A is a schematic of a body of an inflatable implantable tumor treating field system, wherein the body is in a compact configuration. FIG. 10B is a schematic of the body of the inflatable implantable tumor treating field system, wherein the body is in an inflated configuration. FIG. 10C is a schematic of the inflatable implantable tumor treating field system having an electric field-generating assembly within the inflated body.

(12) FIG. 11 is a system for delivering tumor treating fields using the implantable apparatuses and systems as disclosed herein.

(13) FIG. 12 is a schematic of a patient, showing positioning of the implantable apparatuses and systems as disclosed herein.

DETAILED DESCRIPTION

(14) The disclosed system and method may be understood more readily by reference to the following detailed description of particular embodiments and the examples included therein and to the Figures and their previous and following description.

(15) It is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the scope of the present invention which will be limited only by the appended claims.

(16) It must be noted that as used herein and in the appended claims, the singular forms “a”, “an,” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “an electrode” includes one or more of such electrodes, and so forth.

(17) “Optional” or “optionally” means that the subsequently described event, circumstance, or material may or may not occur or be present, and that the description includes instances where the event, circumstance, or material occurs or is present and instances where it does not occur or is not present.

(18) Ranges may be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, also specifically contemplated and considered disclosed is the range from the one particular value and/or to the other particular value unless the context specifically indicates otherwise. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another, specifically contemplated embodiment that should be considered disclosed unless the context specifically indicates otherwise. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint unless the context specifically indicates otherwise. Finally, it should be understood that all of the individual values and sub-ranges of values contained within an explicitly disclosed range are also specifically contemplated and should be considered disclosed unless the context specifically indicates otherwise. The foregoing applies regardless of whether in particular cases some or all of these embodiments are explicitly disclosed.

(19) Optionally, in some aspects, when values are approximated by use of the antecedents “about,” “substantially,” or “generally,” it is contemplated that values within up to 15%, up to 10%, up to 5%, or up to 1% (above or below) of the particularly stated value or characteristic can be included within the scope of those aspects.

(20) Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosed apparatus, system, and method belong. Although any apparatus, systems, and methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present apparatus, system, and method, the particularly useful methods, devices, systems, and materials are as described.

(21) As used herein, the term “patient” refers to a human or animal subject who is in need of

treatment using the disclosed systems and devices.

(22) As used herein, the term “electrode” refers to any structure that permits generation of an electric potential, electric current, or electrical field as further disclosed herein. Optionally, an electrode can comprise a transducer. Optionally, an electrode can comprise a non-insulated portion of a conductive element. In the description herein, any description of a “transducer” or “transducer array” can be applied to an “electrode,” and any description of an “electrode” can be applied to a “transducer” or “transducer array.”

(23) As used herein, the term “stimulation zone” refers to any structural arrangement that permits generation of an electrical field as further disclosed herein. Optionally, a stimulation zone can comprise an electrode and/or a transducer. Optionally, a stimulation zone can comprise a non-insulated portion of a conductive element.

(24) Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude, for example, other additives, components, integers or steps. In particular, in methods stated as comprising one or more steps or operations it is specifically contemplated that each step comprises what is listed (unless that step includes a limiting term such as “consisting of”), meaning that each step is not intended to exclude, for example, other additives, components, integers or steps that are not listed in the step.

(25) Tumor-Treating Fields (TTFields), also referred to herein as alternating electric fields, are established as an anti-mitotic cancer treatment modality because they interfere with proper microtubule assembly during metaphase and eventually destroy the cells during telophase and cytokinesis. The efficacy increases with increasing field strength and the optimal frequency is cancer cell line dependent with 200 kHz being the frequency for which inhibition of glioma cells growth caused by TTFields is highest. For cancer treatment, non-invasive devices were developed with capacitively coupled transducers that are placed directly at the skin region close to the tumor. For patients with Glioblastoma Multiforme (GBM), the most common primary, malignant brain tumor in humans, the system for delivering TTFields therapy is called the OP TUNE™ system (Novocure Ltd.).

(26) Because the effect of TTFields is directional with cells dividing parallel to the field affected more than cells dividing in other directions, and because cells divide in all directions, TTFields are typically delivered through two pairs of transducer arrays that generate perpendicular fields within the treated tumor. More specifically, for the OPTUNE system, one pair of electrodes is located to the left and right (LR) of the tumor, and the other pair of electrodes is located anterior and posterior (AP) to the tumor. Cycling the field between these two directions (i.e., LR and AP) ensures that a maximal range of cell orientations is targeted.

(27) FIG. 11 shows an example apparatus 10 for electrotherapeutic treatment as disclosed herein. Generally, the apparatus 10 may be a portable, battery or power supply operated device that produces alternating electrical fields within the body by means of stimulation zones as disclosed herein (e.g., transducer arrays or electrodes). The apparatus 10 may comprise an electrical field generator 12 and one or more stimulation zones (shown in this exemplary configuration as transducer arrays 14). The apparatus 10 may be configured to generate tumor treating fields (TTFields) (e.g., at 150 kHz) via the electrical field generator 12 and deliver the TTFields to an area of the body through the stimulation zones (e.g., one or more transducer arrays 14 or electrodes). The electrical field generator 12 may be powered by a battery and/or power supply.

(28) As shown in FIG. 11, each transducer array 14 can comprise a plurality of electrodes or transducers 15. In exemplary aspects, the transducers 15 can capacitively couple an AC signal into a subject's body. In further aspects, the transducers 15 can comprise a layer of conductive material, such as a layer of at least one metal (e.g., stainless steel, gold, and/or copper). Additionally, or alternatively, it is contemplated that the transducers 15 can comprise a layer of conductive hydrogel. Exemplary transducers 15 can further comprise ceramic discs, such as described in U.S.

Pat. No. 8,715,203, which is incorporated herein by reference. In additional or alternative aspects, it is contemplated that the transducers **15** can comprise polymer insulating layers and/or other insulating material. Optionally, in exemplary aspects, it is contemplated that one or more of the various layers of the transducers **15** can be flexible and of minimal thickness, thereby allowing easier advancement of the transducer arrays **14** within the body of a subject as further disclosed herein.

(29) The electrical field generator **12** may comprise a processor **16** in communication with a signal generator **18**. The electrical field generator **12** may comprise control software **20** configured for controlling the performance of the processor **16** and the signal generator **18**.

(30) The signal generator **18** may generate one or more electric signals in the shape of waveforms or trains of pulses. The signal generator **18** may be configured to generate an alternating voltage waveform at frequencies in the range from about 50 KHz to about 500 KHz (preferably from about 100 KHz to about 300 KHz) (e.g., the TTFields). The voltages are such that the electrical field intensity in tissue to be treated is typically in the range of about 0.1 V/cm to about 10 V/cm.

(31) One or more outputs **24** of the electrical field generator **12** may be coupled to one or more conductive leads **22** that are attached at one end thereof to the signal generator **18**. The opposite ends of the conductive leads **22** are connected to the one or more stimulation zones (e.g., transducer arrays **14**) that are activated by the electric signals (e.g., waveforms). The conductive leads **22** may comprise standard isolated conductors with a flexible metal shield and may be grounded to prevent the spread of the electrical field generated by the conductive leads **22**. The one or more outputs **24** may be operated sequentially. Output parameters of the signal generator **18** may comprise, for example, an intensity of the field, a frequency of the waves (e.g., treatment frequency), and a maximum allowable temperature of the one or more stimulation zones (e.g., transducer arrays **14**). The output parameters may be set and/or determined by the control software **20** in conjunction with the processor **106**. After determining a desired (e.g., optimal) treatment frequency, the control software **20** may cause the processor **16** to send a control signal to the signal generator **18** that causes the signal generator **18** to output the desired treatment frequency to the one or more stimulation zones (e.g., transducer arrays **14**).

(32) The one or more stimulation zones (e.g., transducer arrays **14**) may be configured in a variety of shapes and positions so as to generate an electrical field of the desired configuration, direction and intensity at a target volume so as to focus treatment. Optionally, the one or more stimulation zones (e.g., transducer arrays **14**) may be configured to deliver two perpendicular field directions through the volume of interest.

(33) Although transducers are conventionally positioned externally on the patient, the present disclosure recognizes that there are benefits to positioning electrodes or transducers within the body of the patient to provide localized electric fields at the site of the tumor. As further described herein, after the electrodes or transducer arrays **14** are selectively positioned within the body of a subject (e.g., in proximity to a target site such as a tumor), an AC voltage can be applied using the electrodes or transducer arrays. A layer of conductive material within the transducers can act as a capacitor's plate, and an insulating layer comprising a polymer, a ceramic material, and/or a coating can act as a capacitor's insulating layer, thereby allowing an AC electric field to be capacitively coupled through the electrodes or transducer arrays **14** into the body of the subject.

(34) Disclosed herein, in various aspects and with reference to FIGS. **1A-1B**, is an implantable tumor treating field (TTField) apparatus **100**. In some aspects, the apparatus **100** can comprise a plurality of elongate elements **104**, each elongate element **104** having a proximal end **106** and a distal end **108**. The plurality of elongate elements **104** can be coupled together at their respective proximal ends **106**. The plurality of elongate elements **104** can comprise at least a first elongate element **104a** and a second elongate element **104b**.

(35) The plurality of elongate elements **104** can be selectively movable about and between a retracted position (FIG. **1B**) and a deployed position (FIG. **1A**). In the deployed position, the

respective distal ends **108** of the first and second elongate elements **104a**, **104b** can be spaced by a first distance **112**. In the retracted position, the respective distal ends **108** of the first and second elongate elements **104a**, **104b** can be spaced by a second distance **114** that is less than the first distance **112**. Exemplary materials for the elongate elements **104** include stainless steel, titanium alloy, cobalt chrome alloy, tantalum, or biocompatible polymer materials. In exemplary aspects, it is contemplated that the elongate elements **104** can comprise a conductive material. For example, in some optional aspects, the elongate elements **104** can comprise conductive thermoplastic materials, which can optionally comprise conductive additives.

(36) In some aspects, the elongate elements can be arranged back-to-back (e.g., in the manner of slats on a hand fan) so that, when in the retracted position, the distal ends of the first and second elements can be aligned in a first transverse dimension (e.g., into the page), and the distal ends can be separated only by the thicknesses of the elongate elements in said first transverse dimension. Thus, for the view shown in FIG. 1B, in some optional aspects, the first elongate element **104a** can entirely cover the second elongate element **104b**. In exemplary aspects, the elongate elements **104** can be pivotably coupled together by a hinge **118** that extends in the first transverse dimension. Optionally, in these aspects, the hinge **118** can comprise conductive material.

(37) In some aspects, the apparatus can comprise an actuator **116** that is configured to pivot the first and second elongate elements **104a**, **104b** about the hinge **118**. The actuator **116** can be, for example, a dial actuator (optionally, a dial actuator coupled to a rotation knob as is known in the art) or other actuator commonly known and used in laparoscopic procedures, such as those used in laparoscopic fan retractors. Optionally, in some exemplary aspects, the actuator **116** can comprise a shaft (not shown) and a catch or lock (not shown) that is coupled to a distal end of the shaft and configured to selectively engage and disengage the proximal ends **106** of the elongate elements **104**, with engagement between the catch/lock and the elongate elements restricting or preventing outward pivotal movement and disengagement permitting outward pivotal movement. It is contemplated that the catch/lock can be selectively engaged with or disengaged from the proximal ends **106** of the elongate elements **104** in response to user-actuation of a button that is mechanically coupled to the catch/lock such that depression of the button causes a release of the catch/lock from engagement with the elongate elements and release of the button causes the catch/lock to return to a locking position. Optionally, the actuator **116** can comprise a spring or other biasing element that biases at least one of the first elongate element **104a** or the second elongate element **104b** toward outward pivotal movement such that upon disengagement of the catch/lock, the elongate elements pivot outwardly. Additionally, or alternatively, in further aspects, the actuator **116** can comprise an actuation cable (not shown) that mechanically couples to at least one of the first elongate element **104a** or the second elongate element **104b**, and a tension applied to the actuation cable can cause the plurality of elongate elements **104** to move to the deployed position. Although described herein as an “actuation cable,” it is contemplated that any elongate element permitting application of tension to the elongate elements through the elongate element (e.g., a cable, a rod, and the like) can be used for this purpose. Additionally, or alternatively, it is contemplated that the actuator **116** can comprise a screw or other rotatable component that is coupled to one of the first or second elongate elements **104a**, **104b** and configured to drive movement of the elongate element about the rotational axis of the screw. In these aspects, it is contemplated that the actuator **116** can further comprise a screwdriver or other rotational actuator that is configured to drive rotation of the screw (or other rotatable component) to drive rotational movement of the elongate element to deploy the elongate elements. Additionally, or alternatively, in still further aspects, the actuator **116** can comprise first and second actuators that are respectively coupled to the first and second elongate elements **104a**, **104b**. For example, in some aspects, it is contemplated that a first rod can be coupled to the first elongate element **104a**, while a second rod can be coupled to the second elongate element **104b**. In these aspects, it is contemplated that the first and second rods can be coupled to the first and second elongate elements such that movement of the proximal ends of the

rods (e.g., rotational movement or axial movement) causes a corresponding translation of the distal ends of the rods, which effects pivotal movement of the elongate elements.

(38) In some optional aspects, the plurality of elongate elements **104** can further comprise one or more intermediate elongate elements **104c**. The intermediate elongate elements **104c** can be positioned between, and coupled to, the first and second elongate elements **104a**, **104b**. For example, webbings or support cables **120** can extend between, and couple to, adjacent elongate elements **104**. In this way, as the first and second elongate elements are moved from the retracted position to the second position, the webbings or support cables **120** can position the intermediate elongate elements **104c** at select positions (e.g., spaced positions) between the first and second elongate elements **104a**, **104b**. Essentially, the first and second elongate elements **104a**, **104b** can be fanned out, thereby spacing the intermediate elongate elements **104c** therebetween. In exemplary aspects, when the elongate elements are in a fully deployed position, it is contemplated that the webbings or support cables **120** can apply sufficient tension to maintain the relative positions of the elongate elements. In these aspects, it is contemplated that the webbings or support cables **120** can comprise flexible materials that do not restrict full retraction of the elongate elements. For example, it is contemplated that the webbings or support cables **120** can flex or be deformed to be accommodated between overlapping portions of adjacent elongate elements as the elongate elements are retracted.

(39) The elongate elements **104** can each comprise one or more stimulation zones **102**, which can optionally be provided as respective locations of one or more transducers or electrodes. In some optional aspects, the stimulation zones can be portions of the elongate elements. For example, optionally, the elongate elements **104** can be generally covered in insulating material (“insulation”), and the stimulation zones **102** can be portions of the elongate elements without insulation. Exemplary insulating materials include, without limitation, polyvinylidene fluoride (PVDF), low-density polyethylene (LDPE), high-density polyethylene (HDPE), fluorinated ethylene propylene (FEP), polyvinyl chloride (PVC), polyolefin-based co-polymers, ethylene acrylic acid-based copolymers, polytetrafluoroethylene (PTFE), or combinations thereof. In further aspects, the stimulation zones **102** can be coupled to the elongate elements **104**. The stimulation zones **102** can optionally be ring-shaped.

(40) As disclosed herein, each stimulation zone can comprise, for example, an electrode **110** or a transducer array. The electrodes **110** can be, for example, conductive or capacitive electrodes. The electrodes can optionally comprise a ceramic material. In further aspects, the electrodes can comprise a metal (e.g., gold, platinum, copper, stainless steel, etc.) or a non-metal (e.g., a polymer such as, for example, polytetrafluoroethylene (ePTFE), polyethylene-co-tetrafluoroethene (ETFE), polyurethanes, silicones, etc.). Optionally, the electrodes can be flexible or rigid. The size and shape of each electrode can be selected based on the desired field produced and the constraints that the respective apparatus provides.

(41) Optionally, each of the stimulation zones can be coupled to a respective electrical lead so that TTFields can be generated between any two select stimulation zones. Accordingly, it is contemplated that each stimulation zone **102** of the plurality of stimulation zones **102** for each elongate element **104** can be independently activated to provide electric fields **130** between itself and any other stimulation zone **102**. For example, a first stimulation zone **102a** and a second stimulation zone **102b** can be activated to generate electric fields therebetween. Thus, each stimulation zones **102** on any elongate element **104** can be independent of the other stimulation zones **102** on the same elongate element **104**. Optionally, the elongate elements can comprise a bundle of electrical leads with exposed portions (without insulation) that function as stimulation zones as disclosed herein.

(42) In some aspects, the implantable TTField apparatus **100** can have an electrical lead **22** (FIG. **11**) coupled thereto and extending therefrom. The electrical lead can comprise a plurality of conductors, with each conductor being in communication with a respective stimulation zone **102**.

In some optional aspects, the electrical lead **22** can have a sufficient rigidity so that a pushing force at the end of the electrical lead opposite the TTField apparatus can effect movement of the TTField apparatus within the body of the patient.

(43) Referring to FIGS. 2A and 2B, in further aspects, an implantable TTField apparatus **200** can comprise a bladder **204**. A plurality of stimulation zones **202** can be coupled to the bladder. In some aspects, the stimulation zones **202** can be portions of the bladder, and the stimulation zones **202** can be coupled to the bladder via integral formation. For example, the bladder **204** can be a conductor (i.e., comprising a conductive material) that is covered or generally covered in insulation (i.e., non-conductive material), and non-insulated portions of the bladder **204** can act as stimulation zones **202**. In some optional aspects, it is contemplated that the bladder can comprise an electrically conductive mesh, conductive nanosheet(s), electrically conductive rubber (e.g., silicone rubber), or a printed conductive layer (with conductive ink).

(44) In further aspects, the bladder **204** can be a non-conductive material, such as, for example, a ceramic or non-conductive polymer, and the stimulation zones **202** can be disposed within, or coupled to an outer surface of, the bladder **204**. Optionally, in exemplary aspects, all or some of the stimulation zones **202** can comprise one or more transducers.

(45) The bladder can be inflated to deploy the stimulation zones **202** from a retracted configuration (FIG. 2C) to a deployed configuration (FIG. 2A). The plurality of stimulation zones **202** can be comprise at least a first stimulation zone **202a** and a second stimulation zone **202b**. When in the retracted configuration, the first and second stimulation zones **202a**, **202b** can be spaced by a first distance **214**. When in the deployed configuration, the first and second stimulation zones **202a**, **202b** can be spaced by a second distance **216** that is greater than the first distance **214**. Optionally, the implantable TTField apparatus **200** can comprise a frame **206** that can be configured to support the bladder and determine the shape of the bladder when the bladder is inflated. For example, the frame can comprise a plurality of webbings **208** that are secured to and extend between portions (e.g., opposing portions) of the bladder so that the webbings restrict expansion of the bladder between the portions of the bladder beyond a select distance. Optionally, the frame can comprise flexible material. Alternatively, the frame can comprise rigid but lightweight materials, such as medical grade metallic or plastic materials.

(46) In some optional aspects, the stimulation zones **202** can be disposed on a plurality of elongate elements **104**. The elongate elements **104** can be each have a proximal end **106** and a distal end **108**. The plurality of elongate elements **104** can comprise at least a first elongate element **104a** and a second elongate element **104b**. The plurality of elongate elements **104** can optionally be coupled together at their respective proximal ends **106** (e.g., via a hinge). The plurality of elongate elements **104** can be coupled to the bladder **204** so that they are deployable via inflation/deflation of the bladder between a retracted position and a deployed position. In the retracted position (FIG. 2C), the respective distal ends **108** of the first and second elongate elements **104a**, **104b** can be spaced by a third distance **210**. When in the deployed position (FIG. 2A), the respective distal ends **108** of the first and second elongate elements **104a**, **104b** can be spaced by a fourth distance **212** that is greater than the third distance. One or more intermediate elongate elements **104c** can be positioned between, and coupled to, the first and second elongate elements **104a**, **104b**. Thus, whereas the elongate elements **104** depicted in FIGS. 1A-1B are deployed by an actuator (e.g., a mechanical actuator) that directly effects movement of the elongate elements, the elongate elements **104** depicted in FIGS. 2A-2C are deployed by inflation of the bladder to which the elongate elements are coupled.

(47) Referring to FIG. 3, in some optional aspects, a implantable TTField apparatus **250** can comprise a plurality of bladders **254** (or a single bladder comprising a plurality of inflatable zones) disposed between adjacent elongate elements **104**. Inflation of the plurality of bladders **254** can bias the adjacent elongate elements **104** away from each other, thereby deploying the stimulation zones from a retracted configuration to a deployed configuration. In some optional aspects, each

bladder **254** can be inflated independently in order to select a spacing between elongate elements **104** that are coupled by each bladder **254**. Thus, in these aspects, it is contemplated that at least one of the bladders **254** (or inflatable zones) can be fluidly isolated from at least other bladder (or inflatable zone). Optionally, it is contemplated that each bladder **254** (or inflatable zone) can be fluidly isolated from every other bladder (or inflatable zone).

(48) Referring to FIGS. **4A** and **4B**, an implantable TTField apparatus **400** can comprise an inflatable body **404** having a longitudinal axis **406**. The body **404** can have a proximal end **408** and a distal end **410** that are spaced along the longitudinal axis **406**. Inflation of the body can cause elongation of the body along the longitudinal axis. For example, in some aspects, the inflatable body can comprise at least one flexible wall **412**. Sufficient pressure within the body can cause the at least one flexible wall **412** to expand to increase the longitudinal length of the body so that the distal end **410** extends in a distal direction (away from the proximal end **408**), while the proximal end stays in the same position along the longitudinal axis **406**. Thus, after inflation, the distal end **410** can be positioned distally relative to the axial location of the distal end prior to inflation. Optionally, the flexible walls can be elastic membranes that stretch. Additionally, or alternatively, the flexible walls can comprise accordion folds to permit expansion and retraction.

(49) Referring to FIGS. **5A** and **5B**, in further optional aspects, prior to inflation, the body **404** can be folded in on itself so that a first portion **414** of the body **404** defines an interior space **416**, and a second portion **418** (e.g., a distal portion) of the body **404** is folded inwardly (e.g., in a proximal direction) to be received within the interior space **416**. The first and second portions **414**, **418** of the body can cooperate to define the length of the body **404**. The second portion **418** can comprise the distal end **410** of the body **404**. As the body is inflated, the body can gradually unfold, thereby pushing the distal end **410** away from the proximal end **408**. In still further aspects, the body **404** can be folded within itself three or more times so that at least a third portion of the body is within an interior space defined by the second portion of the body.

(50) As shown in FIGS. **4A-5B**, the TTField apparatus **400** can comprise a plurality of stimulation zones **402** coupled to the body **404**. The plurality of stimulation zones **402** can comprise a first stimulation zone **402a** and a second stimulation zone **402b**. In some optional aspects, the stimulation zones **402** can be a portion of the inflatable body **404**. For example, the body **404** can be a conductor that is generally covered in insulation, and non-insulated portions of the body **404** can act as stimulation zones **402**. In further aspects, the stimulation zones **402** can be separate components that are coupled to the body **404** (e.g., to an outside of the body or an inside of the body). For example, referring to FIGS. **5A-5B** and **6A-6B**, the stimulation zones **402** can be disposed on a cable **420**. A distal end **422** of the cable **420** can be coupled to the distal end **410** of the body **404**. In exemplary aspects, it is contemplated that the cable **420** can be secured to the body **404** prior to positioning of the TTField apparatus **400** within the body of the patient. Thus, distal movement of the distal end **410** of the body **404** (in response to inflation) can cause distal movement of the distal end **422** of the cable, thereby drawing the cable **420**, and, thus, the stimulation zones **402** outward (away from the proximal end **408**). Optionally, the cable **420** can comprise the electrical leads that couple to the respective stimulation zones. Optionally, in exemplary aspects, all or some of the stimulation zones **402** can comprise one or more transducers.

(51) In some aspects, and as shown in FIGS. **4A** and **4B**, prior to inflation of the inflatable body, the first stimulation zone **402a** can be spaced from the second stimulation zone **402b** by a first distance **426**. Upon inflation of the body **404**, the first stimulation zone **402a** can be spaced from the second stimulation zone **402b** by a second distance **428** that is greater than the first distance **426**. Additionally, it is contemplated that the axial distance between each respective stimulation zone and the proximal end **408** of the body **404** can increase in response to inflation. In some further aspects, and as shown in FIGS. **5A** and **5B**, prior to inflation of the inflatable body, the first stimulation zone **402a** can be spaced from the second stimulation zone **402b** by a first distance **426**. Upon inflation of the body **404**, the first stimulation zone **402a** can remain spaced from the second

conductive zone **402b** by the first distance **426**. However, when the proximal end **408** of the body **404** remains in a fixed position relative to the longitudinal axis, it is understood that the axial distance between each respective stimulation zone and the proximal end of the body will increase in response to inflation. The same characteristics that apply to the embodiment depicted in FIGS. **5A-5B** can also apply to the embodiment depicted in FIGS. **6A-6B**.

(52) Referring to FIG. **7**, in some aspects, at least a portion **430** of the inflatable body **404** can be configured to extend radially outward with respect to the longitudinal axis **406** upon inflation of the inflatable body. Optionally, a plurality of portions **430** of the inflatable body **404** can be configured to extend radially outward with respect to the longitudinal axis **406** of the inflatable body **404**.

Optionally, at least one stimulation zone **402** can be positioned on each of the portions **430** of the body that is configured to extend radially outward. Optionally, each of the portions **430** of the body can be independently inflatable via separate valves or conduits, with each portion **430** being fluidly isolated from other radially extending portions **430**. In some optional aspects, and as shown in FIG. **7**, it is contemplated that the radially extending portions **430** can have a diameter and/or length that is substantially the same as the diameter and/or length of a portion of the inflatable body **404** that extends along the longitudinal axis **406**. In other aspects, it is contemplated that the radially extending portions **430** can have a diameter and/or length that is greater than or less than the diameter and/or length of a portion of the inflatable body **404** that extends along the longitudinal axis **406**. In these aspects, it is contemplated that the diameters and/or lengths can differ by at least 10%, by at least 15%, by at least 20%, by at least 25%, by at least 40%, by at least 50%, or by at least 75%.

(53) By eliminating the need for applying a pushing force (or other mechanical force) to the proximal end of an implanted apparatus to position stimulation zones at a target site, it is contemplated that inflation of the body **404** can minimize damage to tissue as the stimulation zones are positioned at the target site.

(54) Referring to FIGS. **8A** and **8B**, a telescopic implantable TTField apparatus **500** can comprise a first body **504** that is telescopically coupled to a second body **506** so that the first body **506** is slidable with respect to the second body **506** along a longitudinal axis **508**. Optionally, the first and second bodies **504**, **506** can comprise metallic material. For example, an actuator **510** can be configured to slide the first body **504** with respect to the second body **506**. For example, the actuator **510** can comprise a cable **512** that extends through the second body, over a distal end of the second body, and couples to the first body **504** so that retraction of the cable **512** (in a proximal direction) causes distal movement of the first body (see FIG. **8B**). Optionally, the actuator **510** can comprise a plurality of such cables **512**. Alternatively, it is contemplated that the actuator **510** can comprise: a threaded shaft that extends through the second body **506** and is coupled to the first body **504**; and a rotation knob (or other accessible structure) that is rotationally and/or threadedly coupled to the threaded shaft such that rotation of the rotation knob effects a corresponding axial movement of the first body **504**. As another alternative, it is contemplated that the actuator **510** can comprise at least one elongate shaft with an external/accessible portion that extends through the second body **506** and is coupled to the first body **504** such that axial advancement or retraction of the elongate shaft causes a corresponding advancement or retraction of the first body **504**.

Additionally, or alternatively, in various optional aspects, it is contemplated that the actuator **510** can comprise a laparoscopic manipulator that is configured to provide three-dimensional (X-, Y-, and Z-axis control) of devices during a surgical procedure as is known in the art. Although specific examples of the actuator **510** are disclosed herein, it is contemplated that any suitable mechanism for effecting relative axial movement between the first and second bodies can be used.

(55) A plurality of stimulation zones **502** can be disposed along the first body **504**. Optionally, the stimulation zones **502** can be portions of the first body. In further aspects, for example, the stimulation zones can be electrodes that are coupled to individual electrical leads that extend through an interior of the first and second bodies **504**, **506**.

(56) Referring to FIG. 9, in some aspects, an implantable TTField apparatus **550** can have a proximal end **552** and a distal end **554** that are spaced by a length. A plurality of inflatable bodies **556** (e.g., two bodies, three bodies, four bodies, or more) can extend between the proximal end **552** and the distal end **554**. Optionally, in exemplary aspects, the inflatable bodies **556** can be configured for significant axial expansion but only limited radial expansion; optionally, the inflatable bodies **556** can have a maximum inflated diameter that avoids contact with other inflatable bodies. However, embodiments in which a limited amount of contact between inflatable bodies is present are also contemplated. Optionally, the proximal end **552** and the distal end **554** can comprise respective support portions (e.g., circular support portions) that are coupled to the inflatable bodies **556** and formed of flexible material. The inflatable bodies **556** can have respective inflation inputs **558** (e.g., inflation ports) at the proximal end **552**. By selectively inflating one or more of the bodies **556** (optionally, inflating multiple bodies simultaneously) through their corresponding inflation inputs **558**, the distal end **554** can be extended away from the proximal end **552** in a select direction. For example, inflating all bodies simultaneously can cause the distal end to extend away from the proximal end in an axial direction. If all bodies are substantially equally inflated, then the axial direction in which the distal end extends can be a distal or generally distal direction. If a selected body or group of adjacent bodies are more inflated than the remaining bodies, then it is contemplated that the selected body or group of adjacent bodies can dictate the axial direction in which the apparatus extends. For example, in an embodiment having four bodies as shown in FIG. 9, inflating two adjacent bodies on a first transverse side of the TTField apparatus **550** can cause the distal end **554** to extend away from the proximal end **552** in an axial direction moving away from the first transverse side of the TTField apparatus **550**. More particularly, the portions of the two adjacent bodies that extend axially beyond the distal ends of the other bodies are drawn toward the other bodies due to the support portion positioned at the distal ends of the inflatable bodies. A plurality of stimulation zones **560** can be positioned at the distal end and/or along the length of the TTField apparatus **550**. Optionally, at least one stimulation zone **560** can be defined by or coupled to a respective inflatable body **556**. Optionally, the plurality of stimulation zones can comprise at least two stimulation zones that are fixedly secured to or defined by a frame portion at the distal end. In exemplary aspects, it is contemplated that a plurality of fluid (e.g., pressurized air) conduits can be in fluid communication with the respective inflation inputs to permit selective inflation of respective bodies **556**. It is contemplated that a source of air or other fluid can be selectively operated by a clinician to control inflation of the bodies **556**. It is further contemplated that controlled decoupling of the fluid conduits, application of negative pressure, and/or release valves can be used to permit selective deflation of the bodies **556**. In exemplary aspects, it is contemplated that the fluid conduits and TTField apparatus **550** can be advanced or retracted within the body of a patient as a unitary structure using conventional techniques.

(57) Referring to FIGS. 10A-10C, a system **600** can comprise a guide **602** comprising an inflatable body **604** having a longitudinal axis **606**. The inflatable body **604** of the guide **602** can optionally have at least one flexible wall **608** that is configured to expand along the longitudinal axis **606**. In further aspects, the inflatable body can comprise a first portion **610** (e.g., a proximal portion) defining an interior **612** and a second portion **614** (e.g., a distal portion) that, before inflation, can be folded within the interior of the first portion of the body (e.g., in a proximal direction). Optionally, after inflation, the first and second portions of the inflatable body can cooperate to define a length **616** of the inflatable body **602** along the longitudinal axis **606**. In some optional aspects, the guide **602** can comprise a bioresorbable material. Accordingly, after use, the guide **602** can remain in the body of the patient. It is contemplated that the guide can be configured further as described with reference to the body **404** in FIGS. 4A-6B.

(58) An electric field-generating assembly **620** can be configured for receipt within the inflatable body **604** of the guide **602**. The electric field-generating assembly **620** can comprise a cable **622** having a length between a proximal end **624** and a distal end **626**. In some aspects, the cable **622**

can be sufficiently rigid so that pushing on the proximal end **624** of the cable **622** can move the distal end **626** distally into the guide. The interior walls of the inflatable body can guide the electric field-generating assembly **620** toward the distal end of the guide. The electric field-generating assembly **620** can further comprise a plurality of stimulation zones **628**. Optionally, the electric field generating assembly **620** can comprise a cable having a plurality of leads, with each lead extending to a respective stimulation zone.

(59) Positioning the Stimulation Zones

(60) Referring to FIG. **12**, the stimulation apparatuses and systems disclosed herein can be positioned proximate to a target site (e.g., a tumor **132**, FIG. **1A**). For example, for glioblastoma, at least a portion of the implantable TTField apparatus can be positioned in the subdural space. It is contemplated that the stimulation zones can be positioned to generally surround the tumor. In this way, fields emitted therefrom can pass through the tumor. In further aspects, at least one of the stimulation zones can be positioned within or in contact with the tumor. It is further contemplated that the stimulation zones can be positioned so that electric fields traveling between a first pair of stimulation zones can be perpendicular or substantially perpendicular to electric fields traveling between a second pair of stimulation zones. In exemplary aspects, it is contemplated that the intensities of electric fields can be mapped in the target site to determine an optimal placement of the stimulation zones. For example, it is contemplated that Applicant's NOVOTAL system can be used to identify optimal locations for the stimulation zones. As another example, it is contemplated that optimal locations for placement of the stimulation zones can be determined in accordance with the methods disclosed in U.S. Provisional Patent Application No. 63/002,937, entitled "Methods, Systems, and Apparatuses for Assisted Array Placement," which is incorporated herein by reference in its entirety. More particularly, such methods can comprise determining a transducer array map, determining image data associated with a portion of a subject's body, registering the image data to a three-dimensional (3D) model of the portion of the subject's body, wherein the 3D model comprises one or more positions indicated by the transducer array map, and generating composite data comprising the image data and one or more representations of transducer arrays associated with the one or more positions indicated by the transducer array map. In other aspects, such methods can comprise determining a three-dimensional (3D) model of a portion of the subject's body, determining, based on the 3D model, a transducer array map, wherein the transducer array map indicates one or more positions on the 3D model, receiving an image of the portion of the subject's body, determining that the image corresponds to the 3D model, and generating, based on determining that the image corresponds to the 3D model, a composite image comprising the image, the 3D model, and one or more images of transducer arrays associated with the one or more positions.

(61) The stimulation apparatuses and systems disclosed herein can be inserted into the body of the patient **700** from an insertion site **702**. In some optional aspects, it is contemplated that the insertion site can be in the patient's abdomen. In this way, the stimulation apparatus can be less conspicuous and intrusive into the patient's daily life (in comparison to devices implanted near prominent locations on the body). Accordingly, at least a first portion **704** of the apparatus can be disposed at the target site, and another portion **706** of the apparatus can extend to the insertion site. In further aspects, the stimulation apparatus can be inserted into the patient's head or body proximate to the target site.

(62) In some aspects, the disclosed apparatuses can be inserted within the body of the patient using a catheter. However, in some aspects, when a bladder is used as disclosed herein, it is contemplated that the apparatuses can be implanted without the use of a catheter.

(63) In order to determine proper placement, various types of imaging can be used. For example, in some aspects, X-ray imaging can detect the position of the implantable TTField apparatus. In further aspects, various imaging devices can detect conductive material in the apparatus. In still further aspects, radiographic material or other tracking marker(s) can be incorporated into the

apparatus (e.g., at the distal end) for detecting its position within the body of the patient. In further aspects, the apparatus can be configured to emit an electric field that can optionally be different from the tumor-treating fields, and the emitted electric fields can be detected.

(64) Once the stimulation zones have been positioned, the apparatus can be coupled to the electric field generator **12** (FIG. **11**) and can be used to provide TTFields to the target site.

(65) To permit removal of the TTField apparatus, the elongate elements can be retracted.

Optionally, the elongate elements can be retracted via an actuator. Additionally, or alternatively, it is contemplated that movement of the TTField apparatus in a direction away from the patient can cause deployed elongate elements (e.g., fingers) to retract or close.

Positioning of Stimulation Zones for Specific Embodiments

(66) Referring to FIGS. **1A** and **1B**, at least a portion (or, optionally, an entirety) of the implantable TTField apparatus **100** can be inserted in to a body of a patient until a select portion of the apparatus reaches a target site. Once implanted in a desired position, the plurality of elongate elements can be deployed into the deployed position at the target site. For example, the actuator **116** can be actuated to spread the plurality of elongate elements. After use, the actuator can optionally retract the plurality of elongate elements back to the retracted position.

(67) Referring to FIGS. **2A-2C**, at least a portion (or, optionally, an entirety) of the implantable TTField apparatus **200** can be inserted in to a body of a patient until a select portion of the apparatus is in proximity to the target site. Once implanted in a desired position, the bladder **204** can be inflated to deploy the plurality of stimulation zones into desired locations relative to the target site.

(68) Referring to FIG. **3**, at least a portion (or, optionally, an entirety) of the implantable TTField apparatus **250** can be inserted in to a body of a patient until a select portion of the apparatus is in proximity to the target site. Once implanted in a desired position, the bladders **254** can be inflated to deploy the plurality of stimulation zones into desired locations relative to the target site.

(69) Referring to FIGS. **4A-7**, at least a portion (or, optionally, an entirety) of the implantable TTField apparatus **400** can be inserted into the patient. The body **412** can be inflated to position the plurality of conductive zones at desired locations relative to the target site. For example, a gel (e.g., a conductive epoxy or other conductive gel), can be injected into an inlet of the body. In exemplary aspects, the inflatable body can be inserted into the patient using a catheter that guides the inflatable body into a selected position and/or orientation prior to inflation.

(70) Referring to FIG. **8**, the apparatus **500** can be at least partially (or, optionally, entirely) inserted into the body of the patient. The actuator can be actuated to extend the first body with respect to the second body until the stimulation zones are positioned in a desired location with respect to the target site.

(71) Referring to FIG. **9**, the apparatus **550** can be at least partially (or, optionally, entirely) inserted into the body of the patient. Some or all of the inflatable bodies can be inflated (e.g., injecting a biocompatible gel) until the distal end the stimulation zones are positioned in a desired location with respect to the target site. For example, some or all of the inflatable bodies can be incrementally inflated (e.g., all four bodies can be inflated to move the distal end a first distance, then two adjacent bodies can be inflated to move the distal end transversely, then all four bodies can be inflated, etc.) until the desired position of the distal end within the patient is reached.

(72) Referring to FIGS. **10A-10C**, at least a portion of the guide can be inserted into the body of the patient. The body of the guide can be inflated until a distal portion of the body reaches a desired position in the body of the patient. The electric field generating assembly can then be inserted into the guide until the stimulation zones are positioned in a desired location with respect to the target site.

(73) Exemplary Materials

(74) In exemplary aspects, the bladders disclosed herein can comprise one or more of the following materials: silicone, PET, polyvinyl chloride, polypropylene, polyethylene, polyurethanes,

polyamides, polyesters, latex, natural rubber, synthetic rubber, polyether block amides, and elastomers, mixtures, or copolymers thereof. In some aspects, the bladders can comprise multiple layers, each layer comprising one or more of the aforementioned materials.

(75) In exemplary aspects, the stimulation zones disclosed herein can be covered or made with an antimicrobial or hygienic promoting coating such as pMTAC or pDA-g-pMTAC combo, silver, tin, copper, ZnO/Ti spray, DMDC-Q-g-EM hydrogel, platinum, titanium, alloys, stainless steel, cobalt or cobalt-based alloys, cobalt chromium, magnesium alloys, or other material that is biocompatible.

Exemplary Aspects

(76) In view of the described products, systems, and methods and variations thereof, herein below are described certain more particularly described aspects of the invention. These particularly recited aspects should not however be interpreted to have any limiting effect on any different claims containing different or more general teachings described herein, or that the “particular” aspects are somehow limited in some way other than the inherent meanings of the language literally used therein. Aspect 1: An apparatus for generating an electric field, the apparatus comprising: a plurality of elongate elements, the plurality of elongate elements comprising at least a first elongate element and a second elongate element, wherein each elongate element of the plurality of elongate elements has: a proximal end; an opposed distal end; and at least one stimulation zone; wherein the plurality of elongate elements are coupled together at their respective proximal ends, wherein the plurality of elongate elements are selectively moveable about and between a retracted position and a deployed position, wherein, in the retracted position, the respective distal ends of the first and second elongate elements are spaced by a first distance, wherein, in the deployed position, the respective distal ends of the first and second elongate elements are spaced by a second distance that is greater than the first distance. Aspect 2: The apparatus of aspect 1, wherein each stimulation zone comprises an electrode. Aspect 3: The apparatus of aspect 1 or aspect 2, further comprising an actuator that is configured to move the plurality of elongate elements from the retracted position to the deployed position. Aspect 4: The apparatus of aspect 3, wherein the actuator is a mechanical actuator that is mechanically coupled to at least one of the first elongate element or the second elongate element. Aspect 5: The apparatus of aspect 3 or aspect 4, wherein the first elongate element is mechanically coupled to the second elongate element. Aspect 6: The apparatus of aspect 5, further comprising an actuation cable that mechanically couples the first elongate element to the second elongate element, wherein a tension applied to the actuation cable causes the plurality of elongate elements to deploy from the retracted position to the deployed position. Aspect 7: The apparatus of any one of the preceding aspects, wherein the plurality of elongate elements comprises at least one intermediate element that is positioned between, and coupled to, the first and second elongate elements. Aspect 8: The apparatus of any one of the preceding aspects, further comprising at least one cable comprising a plurality of electrical leads, wherein each electrical lead is in electrical communication with a respective stimulation zone of an elongate element of the plurality of elongate elements. Aspect 9: The apparatus of any one of the preceding aspects, wherein at least one elongate element of the plurality of elongate elements comprises a plurality of stimulation zones. Aspect 10: The apparatus of aspect 9, wherein each zone of the plurality of stimulation zones of the plurality of elongate elements is configured to be independently activated to provide electrical fields between itself and any other zone of the plurality of stimulation zones. Aspect 11: An apparatus comprising: a bladder; and a plurality of stimulation zones coupled to the bladder, wherein the plurality of stimulation zones comprise a first stimulation zone and a second stimulation zone, wherein the plurality of stimulation zones are configured to generate an electric field, wherein inflation of the bladder causes the plurality of stimulation zones to deploy from a retracted configuration to a deployed configuration, wherein, when the plurality of stimulation zones are in the retracted configuration, the first and second stimulation zones are spaced by a first distance, wherein, when the plurality of stimulation zones are in the deployed configuration, the first and second stimulation zones are

spaced by a second distance that is greater than the first distance. Aspect 12: The apparatus of aspect 11, further comprising a frame coupled to the bladder, wherein the frame is configured to determine a shape of the bladder when the bladder is inflated. Aspect 13: The apparatus of aspect 11 or aspect 12, further comprising a plurality of elongate elements coupled to the bladder, wherein the plurality of elongate elements comprises at least a first elongate element and a second elongate element, wherein each elongate element of the plurality of elongate elements has: a proximal end; an opposed distal end; and at least one stimulation zone of the plurality of stimulation zones, wherein the plurality of elongate elements are coupled together at their respective proximal ends, wherein the plurality of elongate elements are selectively moveable about and between a retracted position and a deployed position, wherein, in the retracted position, the respective distal ends of the first and second elongate elements are spaced by a third distance, wherein, in the deployed position, the respective distal ends of the first and second elongate elements are spaced by a fourth distance that is greater than the third distance. Aspect 14: The apparatus of aspect 13, wherein the plurality of elongate elements are disposed within the bladder. Aspect 15: An apparatus comprising: an inflatable body having a longitudinal axis; and a plurality of stimulation zones coupled to the inflatable body, wherein the stimulation zones are configured to produce an electric field, and wherein inflation of the body causes elongation of the body along the longitudinal axis. Aspect 16: The apparatus of aspect 15, wherein the inflatable body comprises at least one flexible wall so that a sufficient pressure within the body causes the at least one flexible wall to expand along the longitudinal axis to increase a longitudinal length of the inflatable body. Aspect 17: The apparatus of aspect 15, wherein, prior to inflation the inflatable body, a first portion of the body defines an interior, and a second portion of the body is folded within the interior of the first portion of the body. Aspect 18: The apparatus of aspect 17, wherein, after inflation, the first and second portions of the body cooperate to define a length of the inflatable body along the longitudinal axis. Aspect 19: The apparatus of aspect 15, wherein the plurality of stimulation zones comprises a first stimulation zone and a second stimulation zone, wherein, prior to inflation of the inflatable body, the first stimulation zone is spaced from the second stimulation zone by a first distance, wherein, upon inflation of the inflatable body, the first stimulation zone is spaced from the second stimulation zone by a second distance, wherein the second distance is greater than the first distance. Aspect 20: The apparatus of aspect 15, wherein, the plurality of stimulation zones comprises a first stimulation zone and a second stimulation zone, wherein, prior to inflation of the inflatable body, the first stimulation zone is spaced from the second stimulation zone by a first distance, wherein, upon inflation of the inflatable body, the first stimulation zone remains spaced from the second stimulation zone by the first distance. Aspect 21: The apparatus of aspect 15, wherein the inflatable body comprises at least one portion that is configured to extend radially outwardly with respect to the longitudinal axis upon inflation of the inflatable body, wherein at least one stimulation zone of the plurality of stimulation zones is positioned on the at least one portion that is configured to extend radially outwardly. Aspect 22: The apparatus of aspect 21, wherein the at least one portion that is configured to extend radially outwardly with respect to the longitudinal dimension comprises a plurality of portions that are configured to extend radially outwardly with respect to the longitudinal axis upon inflation of the inflatable body. Aspect 23: An apparatus having a longitudinal axis and comprising: a first body that is telescopically coupled to a second body so that the first body is slidable with respect to the second body along the longitudinal axis; a plurality of stimulation zones disposed along the first body, wherein the stimulation zones are configured to produce an electric field; and an actuator that is configured to slide the first body with respect to the second body. Aspect 24: A system comprising: a guide comprising an inflatable body having a longitudinal axis; an electric field-generating assembly configured for receipt within the inflatable body and comprising: a cable having a length; and a plurality of stimulation zones disposed along the length of the cable, wherein the plurality of stimulation zones are configured to produce an electric field. Aspect 25: The system of aspect 24, wherein the inflatable body comprises at least

one flexible wall so that a sufficient pressure within the inflatable body causes the at least one flexible wall to expand along the longitudinal axis to increase a longitudinal length of the inflatable body. Aspect 26: The system of aspect 24, wherein, prior to inflation the inflatable body, a first portion of the body defines an interior, and a second portion of the inflatable body is folded within the interior of the first portion of the body. Aspect 27: The system of aspect 24, wherein, after inflation, the first and second portions of the inflatable body cooperate to define a length of the inflatable body along the longitudinal axis. Aspect 28: The system of aspect 24, wherein the guide comprises a bioresorbable material. Aspect 29: A method comprising: inserting at least a portion of the apparatus as in any one of aspects 1-10 into a body of a patient until a select portion of the apparatus reaches a target site; and deploying the plurality of elongate elements into the deployed configuration at the target site. Aspect 30: The method of aspect 29 wherein inserting the at least a portion of the apparatus comprises inserting the at least a portion of the apparatus in a patient at an insertion site that is at the patient's abdomen so that the plurality of elongate elements of the apparatus are disposed at the target site and a second portion of the apparatus extends to the insertion site. Aspect 31: The method of aspect 29 or aspect 30, further comprising delivering current through the plurality of stimulation zones to generate an electric field at the target site. Aspect 32: A method comprising: inserting at least a portion of an apparatus as in any one of aspects 11-14 into a body of a patient until the plurality of stimulation zones of the apparatus are in proximity to a target site; and inflating the bladder to deploy the plurality of stimulation zones into desired locations relative to the target site. Aspect 33: The method of aspect 32, further comprising delivering current through the plurality of stimulation zones to generate an electric field at the target site. Aspect 34: A method comprising: inserting at least a portion of an apparatus as in any one of aspects 15-22 into a body of a patient until the plurality of stimulation zones of the apparatus are in proximity to a target site; and inflating the body to deploy the plurality of stimulation zones into the deployed configuration at desired locations relative to the target site. Aspect 35: The method of aspect 34, further comprising delivering current through the plurality of stimulation zones to generate an electric field at the target site. Aspect 36: A method of using the system as in any one of aspects 24-28, the method comprising: inserting the guide into a body of a patient; inflating the body of the guide until the body reaches a desired position in the body of the patient; and inserting the electric field-generating assembly into the guide until the stimulation zones of the electric field-generating assembly are positioned at a desired location with respect to a target site in the body of the patient. Aspect 37: The method of aspect 36, wherein inflating the body comprises injecting a conductive gel into the body.

(77) Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, certain changes and modifications may be practiced within the scope of the appended claims.

Claims

1. An apparatus for generating an electric field to treat a tumor, the apparatus comprising: an electrical signal generator; and a plurality of elongate elements, the plurality of elongate elements comprising at least a first elongate element and a second elongate element, wherein each elongate element of the plurality of elongate elements has: a proximal end; an opposed distal end; and at least one stimulation zone, wherein the plurality of elongate elements are coupled together at their respective proximal ends, wherein the plurality of elongate elements are selectively moveable about and between a retracted position and a deployed position, wherein, in the retracted position, the respective distal ends of the first and second elongate elements are spaced by a first distance, wherein, in the deployed position, the respective distal ends of the first and second elongate elements are spaced by a second distance that is greater than the first distance, wherein the electrical signal generator is electrically coupled to the at least one stimulation zone of each

elongate element of the plurality of elongate elements, and wherein, when the plurality of elongate elements are in the deployed position, the electrical signal generator is configured to generate an alternating voltage waveform at at least one frequency in a range from 50 kHz to 500 kHz to thereby generate tumor treating fields having an intensity from about 0.1 V/cm to about 10 V/cm between the at least one stimulation zone of at least two elongate elements of the plurality of elongate elements.

2. The apparatus of claim 1, wherein each stimulation zone comprises an electrode.

3. The apparatus of claim 1, further comprising an actuator that is configured to move the plurality of elongate elements from the retracted position to the deployed position.

4. The apparatus of claim 3, wherein the actuator is a mechanical actuator that is mechanically coupled to at least one of the first elongate element or the second elongate element.

5. The apparatus of claim 3, wherein the first elongate element is mechanically coupled to the second elongate element.

6. The apparatus of claim 5, further comprising an actuation cable that mechanically couples the first elongate element to the second elongate element, wherein a tension applied to the actuation cable causes the plurality of elongate elements to deploy from the retracted position to the deployed position.

7. The apparatus of claim 1, wherein the plurality of elongate elements comprises at least one intermediate element that is positioned between, and coupled to, the first and second elongate elements.

8. The apparatus of claim 1, further comprising at least one cable comprising a plurality of electrical leads, wherein each electrical lead is in electrical communication with a respective stimulation zone of an elongate element of the plurality of elongate elements.

9. The apparatus of claim 1, wherein at least one elongate element of the plurality of elongate elements comprises a plurality of stimulation zones.

10. The apparatus of claim 9, wherein each zone of the plurality of stimulation zones of the plurality of elongate elements is configured to be independently activated to provide electrical fields between itself and any other zone of the plurality of stimulation zones.

11. A method comprising: inserting at least a portion of the apparatus as in claim 1 into a body of a patient until a select portion of the apparatus reaches a target site, wherein the target site is the tumor; and deploying the plurality of elongate elements into the deployed configuration at the target site.

12. The method of claim 11 wherein inserting the at least a portion of the apparatus comprises inserting the at least a portion of the apparatus in a patient at an insertion site that is at the patient's abdomen so that the plurality of elongate elements of the apparatus are disposed at the target site and a second portion of the apparatus extends to the insertion site.

13. The method of claim 11, further comprising generating, by the electrical signal generator, alternating the voltage waveform to thereby generate the tumor treating fields between the at least one stimulation zone of the at least two elongate elements of the plurality of elongate elements having an intensity from about 0.1 V/cm to about 10 V/cm at the target site.

14. A method comprising: inserting at least a portion of an apparatus as in claim 1 into a body of a patient until the at least one stimulation zone of each elongate element of the plurality of elongate elements of the apparatus are in proximity to a target site, wherein the target site is the tumor; and moving the plurality of elongate elements to the deployed position to deploy each stimulation zone of the at least one stimulation zone of the plurality of elongate elements into desired locations relative to the target site.

15. The method of claim 14, further comprising generating, by the electrical signal generator, the alternating voltage waveform to thereby generate the tumor treating fields between the at least one stimulation zone of the at least two elongate elements of the plurality of elongate elements having an intensity from about 0.1 V/cm to about 10 V/cm at the target site.

16. The apparatus of claim 1, wherein the electrical signal generator is configured to generate the alternating voltage waveform at at least one frequency from 100 kHz to 300 kHz.

17. The apparatus of claim 1, further comprising a webbing or a support cable extending between adjacent elongate elements of the plurality of elongate elements.

18. The apparatus of claim 1, wherein each elongate element of the plurality of elongate elements is elongate along a length, wherein the at least one stimulation zone of a first elongate element of the plurality of elongate elements is elongate along the length of the first elongate element.

19. The apparatus of claim 1, wherein the at least one stimulation zone of a first elongate element of the plurality of elongate elements extends circumferentially around said first elongate element.

20. The apparatus of claim 1, wherein the at least two elements of the plurality of elongate elements, when in the deployed configuration, are configured to be positioned on opposite sides of the tumor.
